



The design of interurban bus networks in city centers

Mireia Roca-Riu^{a,*}, Miquel Estrada^{a,b,1}, César Trapote^{a,1}

^a CENIT, Center for Innovation in Transport, Jordi Girona 29, 2-A, 08034 Barcelona, Spain²

^b UPC, Barcelona Tech., Spain³

ARTICLE INFO

Article history:

Received 14 March 2011

Received in revised form 18 April 2012

Accepted 16 May 2012

Keywords:

Public transit

Transit networks

Transport planning

ABSTRACT

This paper proposes a bilevel formulation for solving the Bus Network Design Problem (BNDP) of interurban services entering a major city. It is focused in interurban services because it is a growing problem in most of major cities, yet new in the literature. The layout of interurban bus routes and the locations of transfer stations in the main city are the key factors to provide a competitive public transportation service to commuters in a metropolitan area. The number of commuters in huge urban concentrations is growing due to the difficulties of living near the city center. The objective function of the first level is defined with the aim of reducing user and agency costs. In the second level the performance of users is addressed. Furthermore, a local search method based on the Tabu Search algorithm was carried out to guide the exploration in the solution domain. The results obtained in a set of test problems have demonstrated that the restart parameters of the algorithm play a significant role in the efficiency of the algorithm. Finally, implementation in the large network of Barcelona (Spain) reduces the total cost by 5% with regard to the present situation.

© 2012 Elsevier Ltd. All rights reserved.

1. Introduction

The key concept of intermodality in public transportation deals with a complementary and systematic design of all transit networks to guarantee a competitive transportation service. A proper location of multimodal stations connecting different networks can decrease travel time and increase accessibility. It is in this context where interurban bus service should have an important role. However, excessive transfers among transit networks are a huge deterrent to passengers as the time spent at stations is perceived very negatively.

The intermodal transportation system presents a hierarchical configuration of services in different geographical scales (urban and interurban). If we focus on interurban bus services, it has to be highlighted that interurban routes run through areas with different infrastructures providing different speed ranges. In interurban areas bus speed can be quite fast and constant. On the contrary, bus travel time in urban areas is limited by the interaction with other vehicles and potential delays at intersections with no priority. The commercial speed of interurban buses drops especially in the city center due to the higher bus stop density, the presence of traffic lights, the effect of right turns, parking operations, and so on. If the city center presents a local transit network with sufficient spatial coverage and transfer stations with interurban bus corridors (intermodal potential), a new paradigm arises: what is the optimal length of interurban routes inside the city center to minimize both user and operator costs?

* Corresponding author. Tel.: +34 934137667; fax: +34 934137675.

E-mail addresses: mireia.roca-riu@upc.edu (M. Roca-Riu), miquel.estrada@upc.edu (M. Estrada), cesar.trapote@upc.edu (C. Trapote).

¹ Tel.: +34 934137667; fax: +34 934137675.

² www.cenit.cat.

³ www.upc.edu.

There is a trade-off between bringing the users closer to their destinations with zero transfers and the loss of speed as interurban vehicles enter the city. In this way, travel time in the local network will affect the optimal location of transfer stations aiming to minimize the total travel time of all interurban trips. The purpose of this paper is to find the optimal location of transfer stops on interurban routes within local networks. This leads to a mathematical formulation of the problem.

The paper is structured as follows: in Section 2 there is a brief breakdown of related contributions in this field, while the model formulation is presented in Section 3. In Section 4, a solving methodology based on Tabu Search algorithm is proposed. In Section 5 the results obtained for several examples are introduced, and finally the conclusions are presented in Section 6.

2. Review

To the best of our knowledge, there is no previous work specifically treating the layout design of interurban routes inside cities. However, several contributions are available concerning different parts of our approach.

The intercity bus design problem refers to connecting different big cities with bus services, while the problem addressed here is the route layout inside a large important city and other towns around it. Yan and Chen (2002) proposed a scheduling model and a solution algorithm for intercity bus carriers. The model is a mixed-integer multi-commodity network flow problem and the algorithm is based on Lagrangian relaxation and a sub-gradient method. Later, Yan et al. (2007) presented a model where a variable market share was considered. In contrast, the model presented in this paper defines a trip-vehicle formulation where the interurban routes are considered and optimized during the whole process.

A bilevel mathematical programming model was proposed by dell'Olio et al. (2006) based on the guidelines of Codina et al. (2003) in order to micro-locate bus stops and optimize frequencies. The authors consider that the bus routes remain fixed and only micro-location of bus stops and frequency determination are the decision variables. Our approach, on the contrary, fixes frequency, avoids micro-location, and considers bus routing.

Several previous works on urban bus route design have been carried out. There is a complete summary in (Kepaptsoglou and Karlaftis, 2009) with many models, approaches and solving methodologies. In (Guihaire and Hao, 2008), a more global review is presented, including, apart from network design, more tactical topics like frequencies setting, timetable developing, bus scheduling and driver scheduling. Two classical models of network design are following mentioned: Ceder and Wilson (1986) proposed an algorithm to find the optimal network design. The algorithm looks at all possible routes between all origin–destination (OD) pairs and complies with service and operational constraints on trip, transfer, and waiting times. Related work by Baaj and Mahmassani (1995) took a heuristic approach which builds one route at a time, following a set of rules defined in a route generation heuristic algorithm (RGA). The program creates routes sequentially, taking into account the OD pair with the highest demand. Once a set of skeleton routes has been generated, another program developed by Baaj and Mahmassani (1990) called TRUST assigns trips according to the OD matrix and determines the frequency and number of buses necessary to satisfy the demand.

The improvement on computational power have led to the emergence of metaheuristics in the last two decades. Pattnaik et al. (1998) solved the route network design problem by following the model of Baaj and Mahmassani (1990) and using a general local search technique based on Genetic Algorithms (GAs). An extension of the algorithm was developed by Tom and Mohan (2003), with new variables and coding scheme. The work of Chien et al. (2001) proposed a GA approach for the design of feeder routes to major intermodal transfer stations but not for the general problem. Ngamchai and Lovell (2003), however, proposed seven specific genetic operators for the same problem and the results showed some improvements. To compare different solving metaheuristics there are the contributions of Fan and Machemehl (2006a,b), Fan and Machemehl (2008). They use a similar model formulation, in the first and second one, they use fixed demand, whether the third one uses variable transit demand. It is interesting to compare the results of different solving metaheuristics used in each: Simulated Annealing, Tabu Search, and Genetic Algorithms, respectively.

The only two references found with real-size applications in bus network design are the following: Zhao and Zeng (2007), who developed a model for urban transit network with the same objective and applied to the Transit Network of Miami-Dade. And the work of Haghani and Banihashemi (2002), however, the problem instead of addressing network design; it focuses on scheduling, multi-depot allocation and route time constraints.

This paper contributes to fill in the existing gap in the literature about the design of interurban networks inside the city center. The formulation is after a bilevel model and the solution methodology an adaptation of the tabu search metaheuristic. The last important contribution of the paper is to provide a real-size application of the model based on the interurban problem inside the city of Barcelona.

3. Problem formulation

The approach considered in this paper is a bilevel formulation of network design for interurban bus routes. In the upper level, the total cost (social, economic, etc.) from the planner's point of view is minimized, and in the lower level the performance of users is determined. We can observe two nested problems: general optimization in the upper level where all objectives are considered, and user optimization in the lower level where the user assignment problem is addressed.

3.1. Sets and indices

Let A be the area of study where the bus stops and transfer stations are set. We consider a graph $G(H; E)$ representing the ground transportation network in the city, where H is the set of nodes and E is the set of links between nodes. The nodes are divided into three groups. The first set I is constituted by the intersections of interurban lines with the border of area A , and is referred to as city gates. The second set J is composed of the centroids of the transportation zones that constitute A . A partition of the area A into homogeneous transportation zones is made for modeling destinations of the transportation. The last set K is composed of the potential interurban stops in the city area. This set comprises all urban bus stops and rail lines in the city. Note that $I \subset K$ since interurban buses can stop at what we consider as the gate of the city. Outside the city there are some nodes denoted G , that are the fixed set of origins where interurban routes will begin and where the demand matrix's generation points are placed. Arcs E are only considered if the street morphology provides minimum characteristics for bus traveling flow: more than one lane or without steep slopes. The graph created will be used to obtain the shortest path between all potential stops with Dijkstra's algorithm before the bus network design method is addressed. The set of routes will be denoted by R .

3.2. Data

Relative to demand, we must use an approximation of the real demand matrix's (d_{ij} , $i \in G$, and $j \in J$) for an interval of time t_0 hours. A passenger between $i \in G$ and $j \in J$ could choose between different alternatives depending on the modes used. Each alternative involves the use of interurban, urban, and/or on-foot phases. The alternative is determined by the stop $k \in K$ where the user leaves the interurban bus. For each of these transfer stops, we can compute a priori each time of the transport chain from the configuration of city services:

- In-vehicle travel time with interurban buses from transfer stop $k \in K$ to transfer stop $l \in K$ ($t_{kl,IU}$).
- Transfer time at the stop k ($t_{k,T}$).
- Waiting time at stop k of the local service ($t_{k,W}$).
- Travel time within the local service, from stop k to a local service stop near j plus access time to j ($t_{kj,TU}$).
- Egress time to destination j ($t_{j,E}$).

Note that subscripts “IU, T, W, TU, E” refer to each possible part of the transport chain: InterUrban, Transfer, Waiting, Urban Transport and Egress, respectively. Also note that this task can be tedious and difficult depending on the complexity of the urban services. For accurate approximations of these data, User Equilibrium Assignment or Stochastic User Assignment in the urban area must be solved.

Finally, as the paper focuses solely on the network layout inside the city borders, the number of different interurban routes r_m is fixed and is considered an input of the problem. Let T_A^r denote the time between the origin of the route r and the gate, f^r its constant frequency, and l_{km} the distance among each pair of the set K (k, m).

3.3. Decision variables

The problem solution of a bus network design is defined by both the interurban routes and the transfer stops κ that will be chosen by users to reach their destinations. Therefore, the key variables are:

- In the Lower level: Transference stops (κ)
- In the Upper level: Nodes and arcs used in R (X)

Variables are defined as follows, $X_{kl}^r = \begin{cases} 1 & \text{if link between } (k, l) \text{ nodes on route } r \text{ is used} \\ 0 & \text{Otherwise} \end{cases}$. These variables will define the configuration of a route as a sequence of stations at which interurban buses stop, the successive nodes $C^r = \{k_1, k_2, \dots, k_{n_r}\}$ from set K which are visited by route r .

As variables representing operator costs (C_O) we use: b the total number of buses needed, b^r the buses needed for each route, T_C^r the time of the route r inside the city, and l^r the length of route r . Regarding users cost (C_U) it is the transference stops (κ) what determines the total user time for each O–D pair. In the next section we formulate cost functions based on these variables.

3.4. Cost functions

The objective is to design the set of routes R inside the city in order to minimize the costs for users and the costs for the bus operator. The formulation of these costs is presented in this section. In the Section 3.6, the two objective functions of the bi-level formulation are built based in these cost functions.

Operator costs will be divided into two types: temporal and distance costs as shown in Eq. (1). The first term of Eq. (1) captures temporal costs. It can be evaluated multiplying the number of buses needed (b) per hours of operation (t_0) and the unit temporal cost (c_t). Respectively, the number of vehicles needed is computed for each route (b^r) and then added in Eq. (2).

The number of vehicles for each route is easily related to the total time of the interurban route and its frequency in Eq. (3), note that $[x]^+$ denotes the smallest integer equal or greater than x . The total time of the interurban route includes: the time of movement of the interurban bus route inside the city (T_C^r), the time of movement outside the city (T_A^r , considered known), plus the rest time for the driver (T_{rest}). Finally, the time of movement of interurban buses inside the city is computed in Eq. (4). We add the times of movement of interurban buses ($t_{ik,IU}$) between two consecutive stops of the route in the subset: $C^r = \{k_2, \dots, k_{n_r}\}$.

The second component of Eq. (1) are distance costs, the component sums all distances of each route (l^r) and multiply it for the unit distance cost (c_d). To compute the distance of each route (l^r), we use a similar sum of Eq. (4) in Eq. (5). We add all distances between two consecutive stops of the route within the subset: $C^r = \{k_2, \dots, k_{n_r}\}$

$$C_O = bc_t t_0 + \sum_{r=1}^{r_m} c_d l^r \quad (1)$$

$$b = \sum_{r \in R} b^r \quad (2)$$

$$b^r = [(T_A^r + T_C^r + T_{rest})f^r]^+ \forall r \in R \quad (3)$$

$$T_C^r = \sum_{k \in C^r = \{k_2, \dots, k_{n_r}\}} t_{kk-1,IU} \forall r \in R \quad (4)$$

$$l^r = \sum_{k \in C^r = \{k_2, \dots, k_{n_r}\}} l_{kk-1} \forall r \in R \quad (5)$$

The user cost is evaluated in terms of perceived travel time. Total travel time from one origin $i \in G$ to one destination $j \in J$ will depend on the transfer stop where users get off interurban buses and board the local transit services. This time will be decomposed into the different phases of the transportation chain and each phase will be weighted with the user perception in each phase (w_T , w_W , w_{TU} , and w_E respectively). The time that users spend on their trips is converted into monetary units with the perceived Value of Time (VoT). Neither access time nor in-vehicle travel time is considered for interurban services out of the city as long as interurban routes are not modified outside the city borders. The total cost of users will be calculated by adding travel times from all origins to all destinations as Eq. (6) shows.

$$C_U = VoT \sum_{i \in G} \sum_{j \in J} d_{ij} [t_{ik,IU} + w_T t_{k,T} + w_W t_{k,W} + w_{TU} t_{kj,TU} + w_E t_{j,E}] \quad (6)$$

3.5. Bilevel formulation

The bilevel model is defined by means of two sets of formulations. On the one hand, in the upper level, the system planner designs interurban bus services, and on the other hand, in the lower level, users choose their best trip in the available network. Eqs. (1)–(11) formulate the upper level problem while Eqs. (12)–(14) determine the demand assignment (lower level).

$$\text{Minimize } f_{obj} = \rho_O C_O + \rho_U C_U \quad (7)$$

subject to : Eqs. (1)–(6)

$$X_{kl}^r - \sum_{n \in K} X_{nk}^r = 0 \quad \forall l \in K \setminus P, \forall k \in K \setminus I \quad (8)$$

$$X_{kl}^r - \sum_{n \in K} X_{nk}^r = 1 \quad \forall l \in K, \forall k \in I \quad (9)$$

$$X_{kl}^r - \sum_{q \in K} X_{lq}^r = 1 \quad \forall l \in P, k \in K \quad (10)$$

$$|P| \leq p \quad (11)$$

$$\text{Minimize } C_U \quad (12)$$

$$\text{subject to : } \kappa \subset K \quad (13)$$

$$C_r \subset K \quad (14)$$

$$X_{kl}^r \in \{0, 1\} \quad (15)$$

3.6. Global objective function: upper level

The aim of the system planner is to minimize the total cost of the transportation system (Eq. (7)) taking into account both bus operator costs and costs incurred by the users. Parameters ρ_O and ρ_U are weights that are normally fixed at 1/2. Recall

that variable X_{kl}^r determines the use of the arc (k, l) in route r . Eqs. (8)–(10) satisfy connectivity between arcs. In particular, Eq. (8) assures that if an arc arrives at stop $k \in K \setminus \{I\}$, there is one and only one arc leaving k . We denote P at the set of final nodes. Eqs. (9) and (10) state the same condition of Eq. (8) involving nodes of subsets I and P . In these nodes the route begins or ends, and Eq. (8) should be adapted. There is one arc leaving (arriving) but there is no continuity backward (forward) yet it is the start (end) of the route.

A strategic hypothesis is that the number $|P| = p$ of terminals where routes can finish their services is fixed a priori in Eq. (11). Terminals are important elements on the network because they have to provide enough space for parking and bus drivers' rest. Finally Eq. (15) states the possible 0–1 values for the integer-decision variables.

3.7. User objective function: lower level

In the lower level a demand assignment model is considered based on the All-or-Nothing (AoN) method. Two assumptions must be made, to apply this method: (i) there is complete information about the generalized cost on all paths and (ii) the perceived value of travel time is the same for all users. The main feature of this assignment is that users always choose their best alternatives in terms of perceived travel time.

AoN would be invalid in local transportation because lots of possible paths with similar travel times can be chosen and congestion effects must be addressed. However, in the demand assignment of interurban users only a few alternatives are available and they can be widely known by users, so the first assumption is reasonable. The second assumption simplifies the model, making solving achievable.

The objective function in the lower level is the term for users in global objective formulation of Eq. (7) and the key variables are transfer stops $\kappa \in K$. Eq. (13) guarantees that transfer stops belong to the subset of potential stops (K), and users leave the interurban routes at stops that belong to the route. Finally, Eq. (14) assures that stops of the routes are from the subset of potential stops (K).

4. Methodology

The proposed methodology to solve the bilevel formulation of this problem is based on Tabu Search (TS), a local search metaheuristic introduced by Glover (1986). TS has a flexible structure which can be adapted to the type of problem in order to adjust better to its attributes. In Fig. 1a, a global structure of the solution algorithm is depicted. At each iteration, some movements of the solution are analyzed and the best configuration is chosen (see Fig. 1b). The demand assignment must be computed by solving the lower level with those routes determined in the upper level, in order to obtain the new value of the objective function.

Contrary to classical descent methods, the current solution may deteriorate from one iteration to the next to avoid paths that have already been investigated. Before the new iteration begins, long-term memory, *tabu* status, and solution unfeasibility are checked. The *tabu* status is designed to prevent cycling but can be overridden if certain conditions are met; this is called the aspiration criterion. Long-term memory and unfeasibility are techniques often employed to intensify or diversify the search process. It is noteworthy that the form of the TS algorithm has been adapted to the bilevel formulation of this problem as lower and upper levels are considered at each iteration.

4.1. General searching process

An initial solution consists of the interurban routes definition and the users' transfer stop choices. The transition from the current solution x to the new solution $x' \in N(x)$ in the neighborhood can be obtained by performing some movements. At each iteration we select one stop of one route to be modified and we evaluate the movements over a number of nv closest stops to the stop being modified. The neighborhood of the solution is built evaluating all the movements in the nv -closest stops. Then, the best solution of the neighborhood is selected. The movements defined for this problem are the following (see Fig. 1b):

- Change the destination of one route to another stop $k \in K$ of the neighborhood (Mov1).
- Delete the destination of one route, making the route shorter and changing the final destination of the previous one (Mov2).
- Add a new stop from the neighborhood at the end of one route, making the route larger and changing the final destination to the new destination (Mov3).
- Change an intermediate stop on one route to another $k \in K$ of the neighborhood (Mov4).

When a single stop is removed from a particular route at iteration t , the *mortality tabu* value (Θ) prevents this stop allocation being repeated until iteration $t + \Theta$ is reached. On the other hand, the *life tabu* value (θ) avoids removing those stops included in a defined route of the current solution for the next θ iterations. In order for both *tabu* values to be effective, the condition $\Theta > \theta$ must be verified.

Every solution considered during the search process is evaluated by Eq. (16), which tries to determine its goodness. The algorithm tries to find a solution that minimizes the value of Eq. (16).

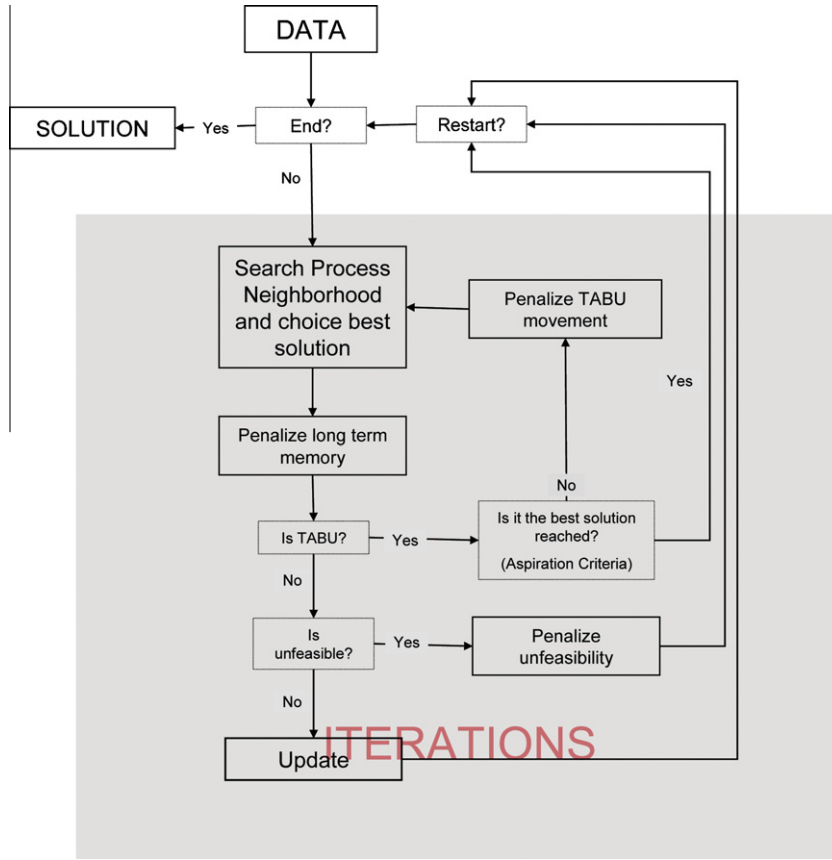


Fig. 1a. Scheme of the iterative method.

$$f(x) = C(x) + \gamma g(x) + \alpha p(x) \quad (16)$$

This function is similar to the one proposed by Cordeau and Laporte (2003) and Scheuerer (2006) and consists of three terms: $C(x)$ represents the real cost of the current solution defined in Eq. (7). In order to diversify the search, long-term frequency memory will be considered, penalizing those movements that are historically more frequently used to lead the exploration to new domains. This penalization is expressed in the evaluation solution function as a term added to the cost function, $g(x)$, which is defined by Eq. (17). A constant factor γ is used to adjust the intensity of the diversification.

$$g(x) = (|K||R|nv)^{1/2} C(x) \rho_{jr} / \lambda \quad (17)$$

where is the $|K|$: size of set K and $|R|$: size of set R the nv : number of candidates, size of neighborhood the ρ_{jr} : number of times that attribute j is included in route r (transition frequency) the λ : number of the current iteration.

The *oscillation criterion* is represented by the term $p(x)$ in Eq. (16), which is proportional to the unfeasibility of the solution. When any constraint is violated, the method continues with unfeasible solutions but its cost is penalized. Unfeasibility could only be reached by the restriction defined in Eq. (11), where $p(x)$ is equal to the number of unwanted destinations. A self-adjusted parameter α is used to control the penalization. The parameter is constrained in order to satisfy some limits and is determined in Eq. (18), where k_1 and k_2 refer to the number of feasible and unfeasible iterations respectively. The auto-regulation of parameter α is from Cordeau et al. (1997). Note that if the solution x is feasible, then $p(x) = 0$.

$$\alpha = \max \left(\min((1 + \delta)^{k_1} / (1 + \delta)^{k_2}, 100), 0.1 \right) \quad (18)$$

To inhibit penalizations when the new solution is the best one found during the search process, functions $f(x)$ and $g(x)$ are zero and the *tabu* list is ignored. This is called the *aspiration criterion* and, despite its simplicity, it provides very good results.

4.2. Solving the lower level problem

The passenger assignment problem must be solved in each iteration with the lower level model. This means determining where users are going to leave the interurban bus network. To solve this problem, a simple heuristic with some final

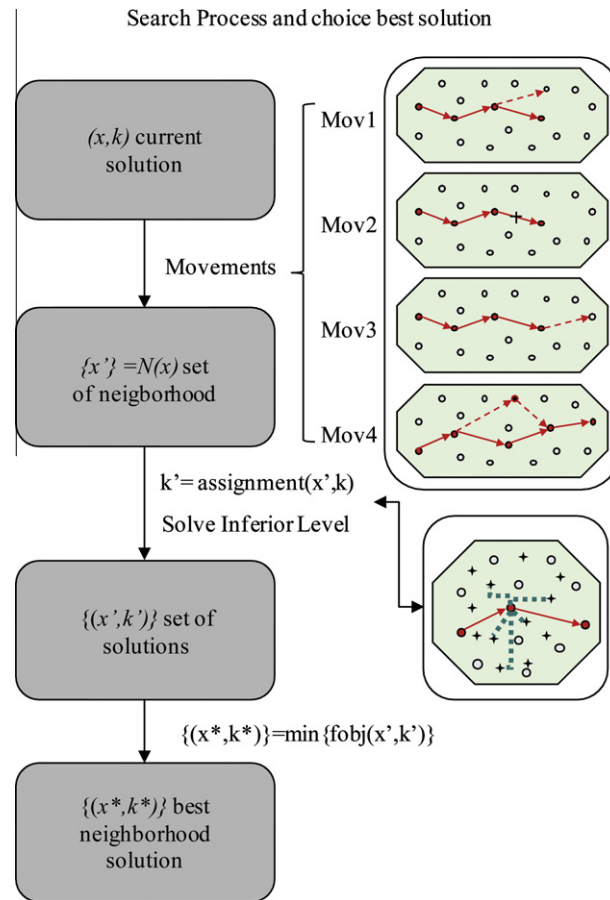


Fig. 1b. Search process.

reassignments is employed. Following the AoN assignment, the shortest path for every user will be computed. If capacity constraints are active, the assignment is made taking into account only those routes with available seats. The heuristic is summarized as follows. We let L be the set of all OD pairs.

Setp I. Consider $l \in L$.

Setp II. Search the best route, excluding full to capacity routes for l . Remove l from L .

Setp III. If L is not an empty set, return to Step I.

Setp IV. Consider possible passenger transfers from full to capacity routes to available ones in order to decrease user-perceived time.

Although this heuristic needs intensive computational time, the demand assignment made in the previous iteration can be used to reduce the number of operations. Given the previous passenger assignment, solving the lower level only entails reassigning the affected users, who represent a low percentage of the total demand.

5. Results and discussion

A set of two benchmark instances has been generated in order to calibrate the algorithm in a typical transit network layout of small cities. These problems have been created following grid or radial network layouts in the city. The potential inter-urban stops are located at street intersections. The average distance between them is approximately 500 m and there are four main city gates for the radial network and two gates for the grid one. The demand is balanced-distributed from five origins to nine centroids in the grid network and equally distributed from three origins to 15 centroids in the radial one. The local transit network is constituted by two bus routes in the former network and two subway lines, two bus routes, and one tram in the latter network. There are six and three interurban routes respectively, with frequencies ranging from two to six buses per hour.

Additionally, a real test instance has been considered in the city of Barcelona that presents a large interurban bus network and a huge number of potential transfer stops to local transit services. This problem will be used to identify the algorithm efficiency with regard to current interurban bus configurations in a large network. The unit distance and temporal costs for one bus are respectively $c_d = 0.667$ €/km and $c_t = 55.81$ €/h based on data from CENIT (2010). These values are taken from the accounts of the major bus operator in the city of Barcelona.

In the case of Barcelona city some adaptations were necessary to adjust the available real data and simplify the bus network. Instead of using all local stops as potential interurban stops we selected local stops that fulfill the following conditions: the existence of a railway interchange or an interchange with more than four lines of local bus services, only one stop at less than 400 m, and zero stops at less than 200 m. The reduction of the number of potential interurban stops follows two objectives. On the one hand, with less number of possible stops the algorithm's search is faster. On the other hand, the idea of the paper is to decide the path of the routes and the stops, but microlocation of the stops is not addressed. The graph of streets is formed from the streets connecting potential interurban bus stops. The current available transit network has nine subway lines, six tram lines, and 76 local bus routes. There are only six big arterial access corridors to the city and one is not actually used because there is a toll system. Thirty-seven interurban routes are considered from 31 different origins in the local area and some big centers of populations around 60 km away. The different destinations inside Barcelona city that are used are extracted from a mobility survey (IDESCAT, 2001). The partitions divide Barcelona into 200 zones of transportation and their corresponding gravity centers. The demand matrix is thought to consider the total mobility during a two-hour period in the morning rush hour.

5.1. Adjustment of the parameters

The parameters are conceived to control three different features of the algorithm: to weight the different penalizations of the objective function, to restrict the search space in each iteration, and to decide when to stop the algorithm. The adjustment of these parameters was tested in the benchmark instances. Several experiments with the same parameters were executed to smooth the randomness of the algorithm. Then, an interval for each was recommended for our problem in relation to the dimension of the problem.

The adjustment of the parameters *tabu tenure* (life and mortality, θ and Θ) is shown in Fig. 2. Life *tabu* should be $\theta > 2$, but there is no further distinction between the other values. As regards mortality *tabu*, some specific values work correctly but they do not show a general tendency. It could be considered that $\Theta > 18$ provides good solutions with low variation among different replications.

The oscillation criterion, which allows infeasible iterations, is controlled by α and δ parameters. Several tests were performed as a result of which the former parameters were fixed at 1 and 0.2 respectively. These values allow some infeasible solutions but limit an extensive search in infeasible domains. In Fig. 3, each infeasible iteration is depicted by an asterisk (*). It can be observed that when the procedure is near to a local optimum some infeasible iterations are made before the procedure becomes incapable of continuing to improve.

Brandao (2006) suggests using a restart value relative to the square root of N_T , being N_T the dimension of the problem. In particular, $N_x = m_x \sqrt{N_T}$ with $m_1 \in [40, 1300]$, $m_2 = 18$, and $m_3 = 28$ and we use $|K|$ as problem dimension. An example of the effect of restart criteria is shown in Fig. 4, where the use of the restart value leads to a better solution.

Parameter γ was tested for several values between 10^2 and 10^{-5} . If γ is large the effect of long-term memory is too strong and inhibits good solutions. On the contrary if γ is too small, the effect is not appreciable (see Fig. 5). Moreover, if γ is large the evolution of the objective function with iterations varies highly from one iteration to the next and tends to increase the objective function. Therefore, the recommended value of γ is 0.05.

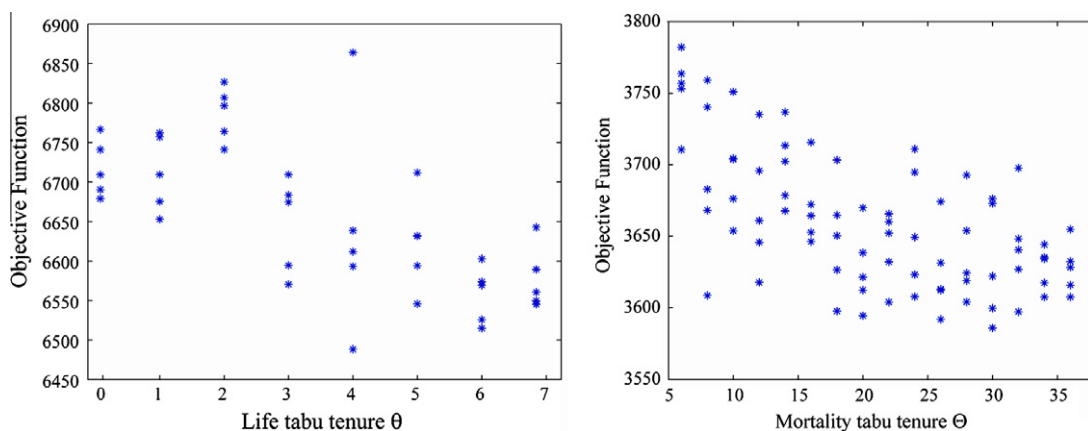


Fig. 2. Calibration of life and mortality *tabu* versus objective function value.

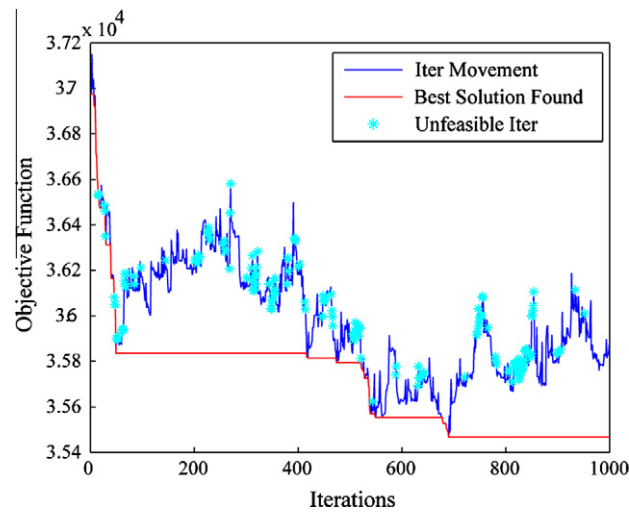


Fig. 3. Effect of unfeasible iterations.

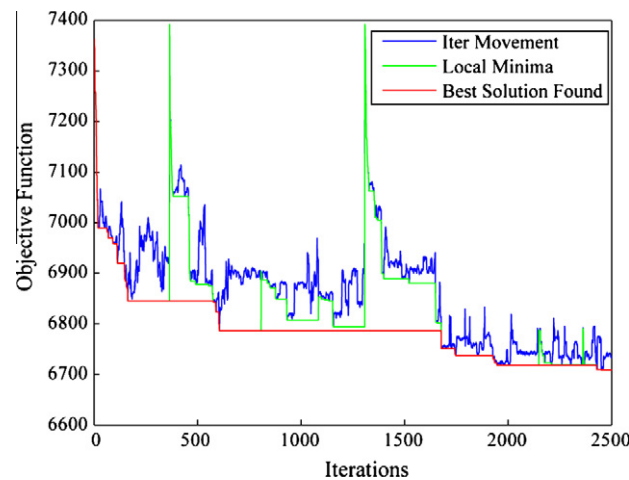


Fig. 4. Effect of restart criteria after some iterations without improvements.

The determination of the number of stops evaluated in the neighborhood (nv) is not only related with obtaining good solutions, but also related with the computational time necessary to finish the execution. For that reason, there is a trade-off between good solution and reasonable computational time. There is an understandable tendency of increasing computational time when increasing the size of the neighborhood explored, because a deeper exploration entails more assignments and computing of the objective function. To adjust this parameter we will use the relative size of neighborhood, meaning $(nv/|K|)$ instead of directly nv . The reason is to adapt better the parameter to the problem size. Fig. 6 shows the evolution of the total computational time (cyan color) with different neighborhood sizes, while Fig. 7 shows the values of the objective function obtained with this neighborhood size. From this analysis we recommend using neighborhood sizes between $1/10$ and $1/8$ in relation to the size of the problem where the best values of the objective function are reached.

5.2. Computational time and bilevel performance

The whole methodology has been implemented in MATLAB code and run on an Intel Core 2 Quad 2.4 GHz under Windows XP. The algorithm performance has been analyzed and the detailed computational times in different runs of the grid network are provided. Different runs for several dimensions of the neighborhood are compared in Fig. 6. It is clear that nearly half of the computing time is used to carry out the demand assignment and the other half to compute the objective function. This part of the algorithm is repeated many times; therefore, it is crucial to reduce the number of operations in both parts. In test networks the calculation time for the values used was 76 s, corresponding to 0.03 s per iteration. In the Barcelona network, the computational time was 5618.41 s (1.56 h) for 5000 iterations.

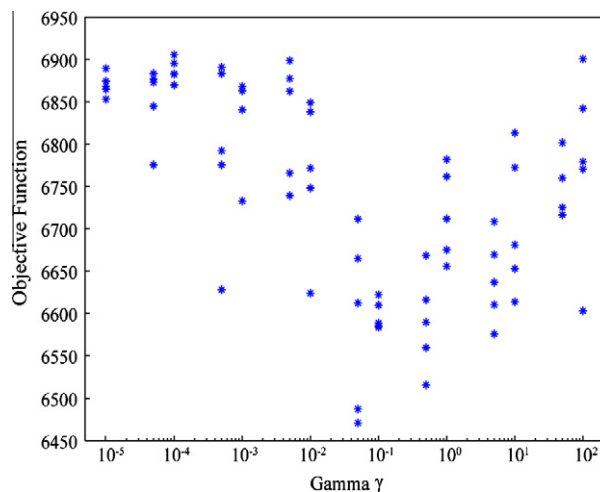


Fig. 5. Gamma effect.

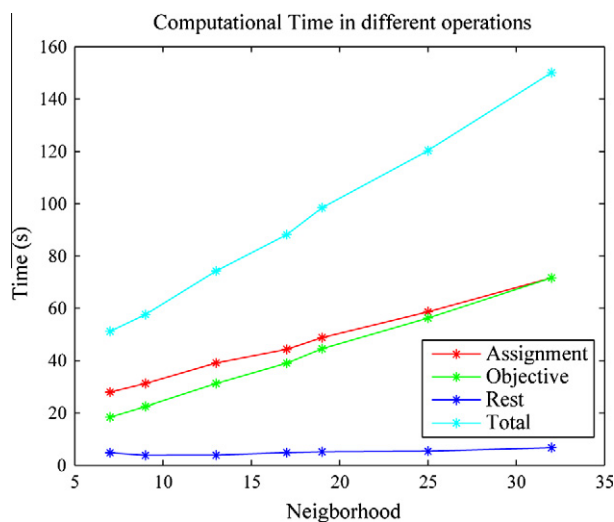


Fig. 6. Computational time in different operations and for different values of objective functions.

5.3. Discussion

Finally, the solutions obtained for each network are summarized in Table 1. The speed of interurban buses inside the city is 18 km/h, and the speed in the urban transit network is 12 km/h for local buses, 22 km/h for trams, and 25 km/h for the metro. For the two theoretical networks there is a comparison between an initial solution and the results obtained. The initial solution consists of medium-sized routes in the grid network and short routes in the radial one. In the case of Barcelona, the present situation is used as the initial solution and we can quantify the improvements that could be made. Regarding operator costs, we present the total hourly cost for the operator, detailing the number of buses needed and the mean route length inside the city. For users we show: firstly, the total monetary cost that their time inside the city represents; secondly, the unitary time, which is the average time that a passenger spends in the city system between city entrance and destination; and, finally, the percentage of that time that users spend on the interurban bus. Note that present interurban demand will always be satisfied with the routes proposed here, yet the route modification is only inside the city.

The robustness of the methodology was evaluated in some experiments. The results show that the best solution does not differ too much from the others obtained with different evaluations and from different initial solutions. The maximum differences in the objective function were 3.01% for the grid network, 1% for the radial network, and 1.3% for the Barcelona network. However, the structure of the routes and their lengths were generally the same.

Another result worth mentioning is the evolution of the optimum interurban network with different urban speeds. The results of the methodology with different urban speed produce the following results. The decrement of urban speed to

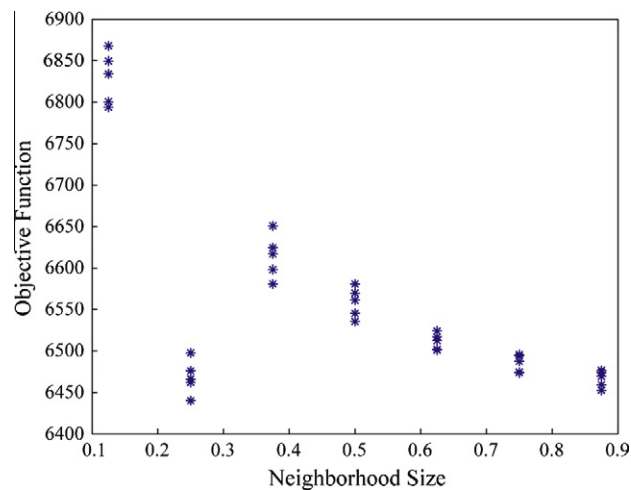


Fig. 7. Neighborhood size and objective function.

Table 1

Results for grid, radial, and Barcelona networks.

Total cost (€)	Operator cost			User cost		
	Cost (€)	Mean route length (km)	Number of buses	Cost (€)	Unitary time (min)	Interurban time (%)
<i>Grid network</i>						
Initial	7391.56	702.74	12	6688.82	19.2	
Best found	6500.7	719.75	12	5781.22	16.8	
Difference	13.7%	−2.36%	+1.42	15.7%	−2.4	–
<i>Radial network</i>						
Initial	5200.98	298.43	5	4902.55	20.4	14.9%
Best found	4461.93	361.35	6	4100.58	16.8	29.7%
Difference	16.56%	−17.4%	+1.18	19.55%	−3.6	–
<i>Barcelona</i>						
Present	36985.59	9297.96	81	27687.63	52.2	16%
Best found	34895.12	7843.58	68	27051.54	51	18%
Difference	5.56%	18.54%	+0.19	2.35%	−1.2	–

14 km/h involves savings of 3.76% for the operator but an increment of 6% in the user's costs. The increment of urban speed to 22 km/h only increases the operator's costs by 0.62% but reduces the user's costs by 4.6%. This data makes clear that the increase in urban speed provides an outstanding gain in user service without a significant increase in operator costs. In contrast, a decrease in speed saves little money for the operator. Therefore, we have to conclude that urban speed is a key parameter to adjust the model to reality.

6. Conclusions

The model is capable of representing reality and modeling different route alternatives for the user. The bilevel choice has proved to be adequate in taking into account global and user optimization. The objective function used, while decreasing costs, is also a good guide to improve other variables like commercial speed or route coverage. Moreover, the model can be adapted to cope with different arc speeds or different local network designs. On the other hand, the methodology proved to be efficient in all networks, and it is suitable to become a general tool capable of solving problems related to minimal changes. Practical results show a decrease in global costs by 5–10% and decreases in both operator and user costs.

Acknowledgments

This research was supported by the Science and Education Ministry (MEC) of Spain with the Grant TRA2005-09305-C02-01 (INTERBUS project) and by the Centro de Estudios y Experimentación de Obras Públicas (CEDEX) of Spain with the grant PT2006-027-06IAPP.

Appendix A

Type	Name	Description
Set/indices	A	Area of study
	E	Links representing streets
	G	Origins outside the city
	$G(H; E)$	Graph representing street configuration
	H	Nodes $H = \text{SetUnion}(I, J, K)$
	I	City gates
	J	Destination centroids inside city
	K	Potential users stops
	P	Set of final stops
	R	Set of routes
Data	c_t	Unitary cost for time
	c_d	Unitary cost for traveled distance
	d_{ij}	Demand between nodes i, j
	l_{kl}	Distance between stops k and l
	t_0	Number of hours considered in the service
	f^r	Constant frequency of route r
	p	Maximum number of final stops
	r_m	Number of routes
	T_A^r	Time between origin of the route and city gate
	T_{rest}	Drivers rest time
	$t_{ik, IU}$	Time with interurban service between city gate i and stop k . Subscript “IU” for InterUrban
	$t_{k, T}$	Transfer time in stop k . Subscript “T” for Transfer
	$t_{k, W}$	Waiting time in stop k . Subscript “W” for Waiting
	$t_{kj, TU}$	Time with urban service between stop k and destination j . Subscript “TU” for Urban Transport
	$t_{j, E}$	Egress time to destination j . Subscript “E” for Egress
	w_T	Weights of user perception for transfer time
	w_W	Weights of user perception for waiting time
	w_{TU}	Weights of user perception of urban transport time
	w_E	Weights of user perception of egress time
	ρ_O	Weights of operator cost
	ρ_U	Weights of user cost
Variables	b	Total number of buses
	b^r	Buses for route r
	l^r	Distance of route r
	T_C^r	Time of route r
	X	Arcs used in routes
	κ	Transference Stops
Algorithm parameters	nv	Number of stops evaluated in the neighborhood
	Θ	Mortality tabu, number of iterations that avoid undoing a movement previously done
	θ	Life tabu, number of iterations that avoid re-doing a movement previously done
	γ	Weight of long term memory of the function that guides the search process
	α	Weight of unfeasibility term of the function that guides the search process
	δ	Parameter used to define α
Other algorithm notation	x	Solution of the problem
	$N(x)$	Set of neighborhood solutions of solution x
	t	Algorithm iteration
	$g(x)$	Long term memory term of the search function
	$p(x)$	Unfeasibility term of the search function
	$C(x)$	Cost of the objective function of solution x
	$f(x)$	Search function of the algorithm
	ρ_{jr}	Number of times that attribute j is included in route r (transition frequency)

Appendix A (continued)

Type	Name	Description
	λ	Iterator counter
	k_1, k_2	Feasible and unfeasible iterator counter
	L	List of all OD pairs of demand matrix
	N_T	Dimension of a problem
	N_x	Number of iterations before a restart in the algorithm
	m_x	Factor that determines the test of different restart values

References

- Baaj, M.H., Mahmassani, H., 1995. Hybrid route generation heuristic algorithm for the design of transit networks. *Transportation Research Part C: Emerging Technologies* 3 (1), 31–50.
- Baaj, M.H., Mahmassani, H.S., 1990. TRUST: a LISP program for the analysis of transit route configurations. *Transportation Research Record* 1283, 125–135.
- Brandao, J., 2006. A new tabu search algorithm for the vehicle routing problem with backhauls. *European Journal of Operational Research* 173 (2), 540–555.
- Ceder, A., Wilson, N.H.M., 1986. Bus network design. *Transportation Research Part B: Methodological* 20 (4), 331–344.
- CENIT, 2010. Xarxa de Transport Públic de Barcelona. Bases per a la definició d'un nou model integrat dels serveis de superfície. (Public Transport Network of Barcelona. Foundations for a new integrated model of surface transport). Barcelona, Spain.
- Chien, S. et al, 2001. Genetic algorithm approach for transit route planning and design. *Journal of Transportation Engineering* 127 (3), 200–207.
- Codina, E. et al, 2003. Programación matemática binivel en planificación del transporte urbano: estado del arte y nuevos modelos. *Estudios De Construcción y Transportes* 99, 127–149.
- Cordeau, J. et al, 1997. A tabu search heuristic for periodic and multi-depot vehicle routing problems. *Networks* 30 (2), 105–119.
- Cordeau, J., Laporte, G., 2003. A tabu search heuristic for the static multi-vehicle dial-a-ride problem. *Transportation Research Part B: Methodological* 37 (6), 579–594.
- dell'Olivo, L. et al, 2006. Bi-level mathematical programming model for locating bus stops and optimizing frequencies. *Transportation Research Record* 1971 (1), 23–31.
- Fan, W., Machemehl, R.B., 2008. A tabu search based heuristic method for the transit route network design problem. In: Fan, W., Machemehl, R.B. (Eds.), *Computer-Aided Systems in Public Transport*. Springer, Berlin, Heidelberg, pp. 387–408.
- Fan, W., Machemehl, R.B., 2006a. Optimal transit route network design problem with variable transit demand: genetic algorithm approach. *Journal of Transportation Engineering* 132 (1), 40–51.
- Fan, W., Machemehl, R.B., 2006b. Using a simulated annealing algorithm to solve the transit route network design problem. *Journal of Transportation Engineering* 132 (2), 122–132.
- Glover, F., 1986. Future paths for integer programming and links to artificial intelligence. *Computers and Operations Research* 13 (5), 533–549.
- Guihaire, V., Hao, J., 2008. Transit network design and scheduling: a global review. *Transportation Research Part A: Policy and Practice* 42 (10), 1251–1273.
- Haghani, A., Banihashemi, M., 2002. Heuristic approaches for solving large-scale bus transit vehicle scheduling problem with route time constraints. *Transportation Research Part A: Policy and Practice* 36 (4), 309–333.
- IDESCAT, 2001. Encuesta de Movilidad Obligada.
- Kepaptsoglou, K., Karlaftis, M., 2009. Transit route network design problem: review. *Journal of Transportation Engineering* 135 (8), 491–505.
- Ngamchai, S., Lovell, D.J., 2003. Optimal time transfer in bus transit route network design using a genetic algorithm. *Journal of Transportation Engineering* 129 (5), 510–521.
- Pattanaik, S.B. et al, 1998. Urban bus transit route network design using genetic algorithm. *Journal of Transportation Engineering* 124 (4), 368–375.
- Scheuerer, S., 2006. A tabu search heuristic for the truck and trailer routing problem. *Computers & Operations Research* 33 (4), 894–909.
- Tom, V.M., Mohan, S., 2003. Transit route network design using frequency coded genetic algorithm. *Journal of Transportation Engineering* 129 (2), 186–195.
- Yan, S., Chen, H., 2002. A scheduling model and a solution algorithm for inter-city bus carriers. *Transportation Research Part A: Policy and Practice* 36 (9), 805–825.
- Yan, S. et al, 2007. An intercity bus scheduling model incorporating variable market share. *IEEE Transactions on System, Man and Cybernetics, Part A: Systems and Humans* 37 (6), 921–932.
- Zhao, F., Zeng, X., 2007. Optimization of user and operator cost for large-scale transit network. *Journal of Transportation Engineering* 133 (4), 240–251.